Solver Trace Ablation Reveals Reward Hacking in GRPO-Trained Formalizers: Meta-Level Goodhart's Law in Structural Proxy Metrics

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Abstract

Reinforcement learning from execution feedback (GRPO) trains neurosymbolic formalizers to high solve rates, but standard evaluation via prove rate alone misses structural degeneration. We propose *Solver Trace Ablation* (STA), a diagnostic protocol that derives the *Intensional Structural Reward* (ISR) through leave-one-out component necessity testing on solver inputs. Across first-order logic (FOL) and Prolog formalizations with 3 random seeds and up to 1000 training steps, we show that (1) ISR detects reward hacking invisible to prove rate—ISR collapses from 0.64 to 0.003 while prove rate rises from 32% to 96%; (2) using ISR as a training reward induces *meta-level Goodhart's Law*: at low dose (β=0.1), models game ISR itself via conclusion duplication (H7 rate 64.5% vs. 0% in random control), while the apparent structural improvement vanishes under premise-count matching (1.06×, no real advantage); at high dose (β=0.3), classic Goodhart collapse occurs (prove rate 18% vs. 48% for random reward); and (3) ISR collapse replicates across logic types (Prolog exec-only 3-seed ISR 0.008–0.081), providing preliminary cross-logic evidence with consistent but heterogeneous hacking patterns. We identify 6 distinct hacking strategies, confirm via SFT baseline that hacking is RL-specific (92.9% genuine vs. 12.9% under GRPO), and document ISR's fundamental asymmetry: reliable as observer, unreliable as optimizer.

1 Introduction

Group Relative Policy Optimization (GRPO) with execution-only rewards has emerged as a dominant paradigm for training neurosymbolic formalizers that translate natural language into formal logic for automated theorem proving. These systems are typically evaluated by *prove rate* (PR)—the fraction of problems for which the generated formalization leads to a successful proof. However, high prove rate does not guarantee structural quality: a model may achieve near-perfect PR by embedding the conclusion in its premises, rendering the proof trivially valid without genuine reasoning.

We introduce *Solver Trace Ablation* (STA), a diagnostic protocol that applies leave-one-out (LOO) ablation to the components of a solver input (e.g., premises in a FOL formalization or clauses in a Prolog program) and measures the *Intensional Structural Reward* (ISR), defined as the fraction of components whose removal causes the proof to fail. ISR quantifies component necessity: a high ISR indicates that each component contributes to the proof, while a low ISR reveals structural redundancy or degeneration.

Our investigation yields three principal findings:

Claim 1: ISR as Reward Hacking Diagnostic. Execution-only GRPO training achieves PR=93% at 500 steps but ISR collapses from 0.64 (100 steps) to 0.13 (500 steps) to 0.003 (1000 steps). We identify 6 distinct hacking modes through manual annotation of 200 outputs, establishing a taxonomy that enables systematic characterization of structural degeneration.

Claim 2: Meta-Level Goodhart's Law. When ISR is converted from a diagnostic metric into a training reward, models do not produce genuinely better formalizations. At low reward weight (β=0.1), 2 of 3 seeds exhibit ISR-specific gaming via conclusion duplication (H7 rate 96.7% and 86.5%), while the third seed's apparent improvement (2.4× raw ISR) is entirely a premise minimization artifact—premise-count-matched ISR ratio is 1.06×. At high weight (β=0.3), classic Goodhart collapse drives prove rate to 18% (vs. 48% for random reward). A supervised fine-tuning (SFT) baseline confirms that reward hacking is RL-specific: SFT achieves only 14% prove rate but 92.9% genuine formalizations.

Claim 3: Preliminary Cross-Logic Evidence. ISR collapse under execution-only training replicates consistently across FOL and Prolog (Prolog 3-seed ISR 0.008–0.081), providing evidence that the diagnostic protocol transfers across logic types. STA-as-reward Goodhart effects replicate in Prolog with H7-dominant patterns in 2/3 seeds, though cross-seed heterogeneity (H7 range 23%–100%) limits this to suggestive rather than conclusive evidence.

2 Related Work

GRPO for Formal Reasoning. DeepSeek-Prover and related systems use execution-based RL to train formal reasoning agents, evaluating solely on prove rate. EVOM applies scalar execution rewards. These approaches lack structural quality diagnostics.

Reward Hacking and Goodhart's Law. Goodhart's Law—"when a measure becomes a target, it ceases to be a good measure"—is well-documented in RL. Manheim and Garrabrant (2019) taxonomize Goodhart effects into regressional, extremal, causal, and adversarial types. Our *meta-level* Goodhart finding (a diagnostic metric becomes a gaming target) is a specific instance of adversarial Goodhart in the proxy metric design space.

Structural Rewards for Code and Logic. Mellgren et al. (2025) hand-craft Prolog-specific structure penalties. ExecVerify and EGCA derive trace-level rewards for code generation but do not address formal logic. Our LOO-based approach is solver-agnostic and requires no domain-specific engineering.

Feature Attribution and LOO Methods. ISR is conceptually a LOO ablation metric, related to Shapley values and feature importance methods in interpretability. We do not claim novelty in the LOO technique itself; our contribution lies in its application to diagnosing and characterizing reward hacking in GRPO-trained formalizers.

3 Method

3.1 Solver Trace Ablation (STA)

Given a formalization $F = \{c_1, c_2, \ldots, c_n\}$ consisting of n components (e.g., premises in FOL, clauses in Prolog) and a solver $\mathcal{S}$ that returns a binary proof outcome, we define:

$$\text{necessary}(c_i) = \begin{cases} 1 & \text{if } \mathcal{S}(F) = 1 \text{ and } \mathcal{S}(F \setminus \{c_i\}) = 0 \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

The *Intensional Structural Reward* (ISR) is the fraction of necessary components among proved formalizations:

$$\text{ISR}(F) = \frac{1}{n}\sum_{i=1}^{n} \text{necessary}(c_i) \quad \text{if } \mathcal{S}(F) = 1 \tag{2}$$

ISR equals 1 when every component is essential to the proof (maximally tight formalization) and approaches 0 when most components are redundant. For unproved formalizations ($\mathcal{S}(F) = 0$), ISR is undefined; we report *proved-only* ISR throughout.

Computational Cost. STA requires n solver calls per formalization (one per component ablation), yielding $O(n)$ complexity. We evaluate a fast $O(1)$ approximation in Section 4.5.1 and find it insufficient.

3.2 Reward Formulations

We study three reward formulations for GRPO training:

$$r_{\text{exec}} = \mathcal{S}(F) \tag{3}$$

$$r_{\text{ISR}} = r_{\text{exec}} \cdot (1 + \beta \cdot \text{ISR}(F)) \tag{4}$$

$$r_{\text{random}} = r_{\text{exec}} \cdot (1 + \beta \cdot u), \quad u \sim \mathcal{U}(0, 1) \tag{5}$$

where β controls the reward weight. The random reward (Eq. 5) serves as a control to isolate ISR-specific effects from general reward augmentation effects.

3.3 Hacking Taxonomy

Through manual annotation of 200 GRPO outputs across reward conditions, we identify 6 hacking modes:

- **H1** (Conclusion Embedding): The conclusion appears as one of multiple premises, making the proof trivially derivable.
- **H7** (Tautological Collapse): A single premise identical to the conclusion; the "proof" is $p \vdash p$.
- **H3** (Complexity Without Utility): Unnecessary structural complexity that does not contribute to the proof.
- **H4** (Redundant Embedding): Duplicate or near-duplicate components.
- **H5** (Partial Overlap): Components that partially subsume each other.
- **Genuine**: Structurally sound formalizations where all components contribute meaningfully.

4 Experiments

4.1 Setup

We train Qwen-1.5B models using GRPO on two formal reasoning tasks: FOL formalization (evaluated with Prover9) and Prolog formalization (evaluated with SWI-Prolog). All experiments use 3 random seeds (s42, s43, s44) unless noted. We report prove rate (PR) and proved-only ISR as primary metrics, with hacking mode composition from manual annotation.

4.2 Claim 1: ISR as Reward Hacking Diagnostic

4.2.1 ISR Collapse Over Training

Table 1 presents the core finding: ISR collapses monotonically as training progresses, while prove rate increases.

Table 1: ISR collapse trajectory under execution-only GRPO (FOL, seed 42). Prove rate rises while structural integrity degenerates.

Steps	PR	Proved-only ISR	Components	Genuine
100	0.32	0.6423	∼5	∼70%
500	0.93	0.1324	∼8.5	12.9%
1000	0.96	0.0027	18.55	0%

At 100 steps, the model produces relatively few but structurally sound formalizations (PR=32%, ISR=0.64). By 500 steps, prove rate reaches 93% but ISR drops to 0.13, with 87.1% of proved samples containing conclusion embedding (H1). At 1000 steps, ISR collapses to 0.003—a 97.8% reduction from 500 steps—while prove rate barely changes (96%). All 96 proved samples at 1000 steps are classified as H1, with 0% genuine formalizations and a mean component count of 18.55 (bloated, redundant premises).

4.2.2 Cross-Seed Instability

Table 2 shows dramatic cross-seed variation under identical training configurations, itself a key finding.

Table 2: FOL execution-only GRPO at 500 steps across 3 seeds.

Seed	PR	Proved-only ISR	Dominant Mode	Description
s42	0.93	0.1324	H1 (87.1%)	Conclusion embedding
s43	0.14	0.0	Repetition collapse	Training failure
s44	0.98	0.5816	H7 (100%)	Tautological collapse

Seed 42 produces the "standard" hacking pattern (high PR, low ISR, H1 dominance). Seed 43 collapses entirely (PR=14%, ISR=0), demonstrating that single-seed GRPO results are unreliable (σ=47pp across seeds). Seed 44 achieves PR=98% with ISR=0.58, but this is a false positive: 100% of proved samples are H7 (tautological collapse with mean 2.04 components), where high ISR reflects the trivial structure (removing 1 of 2 components necessarily breaks the proof) rather than genuine reasoning quality. This motivates using ISR jointly with component count as a diagnostic.

4.2.3 Hacking Taxonomy

Table 3 presents the 6-mode hacking taxonomy from manual annotation of 200 outputs.

Table 3: Hacking taxonomy: ISR β=0.1 vs. random β=0.1 (seed 42).

Mode	Description	ISR rate	Random rate
H1	Conclusion embedding (multi-premise)	0.519	0.709
H7	Conclusion = sole premise	0.160	0.000
H3	Complexity without utility	0.000	0.038
H4	Redundant embedding	0.037	0.051
H5	Partial overlap	0.025	0.025
Genuine	Structurally sound	0.259	0.177

Total hacking rates are similar (ISR: 74.1%, random: 82.3%), but the composition differs markedly. H7 (tautological collapse) is ISR-specific (16% vs. 0%), while H1 (conclusion embed-

ding) is more prevalent under random reward (70.9% vs. 51.9%). ISR reward shifts the hacking composition from H1 toward H7 without substantially reducing the overall hacking rate.

4.3 Claim 2: Meta-Level Goodhart's Law

4.3.1 ISR β=0.1: Three-Seed Analysis

Table 4 compares ISR reward (β=0.1) against random reward control across all 3 seeds.

Table 4: ISR β=0.1 vs. Random β=0.1, 3-seed comparison (FOL).

Condition	Seed	PR	ISR	H7	H1	Genuine
ISR β=0.1	s42	0.81	0.415	16.0%	51.9%	30.9%
	s43	0.61	0.830	96.7%	0%	∼3%
	s44	0.89	0.902	86.5%	0%	∼13.5%
	Mean±std	0.77±0.14	—	64.5%	17.3%	∼16%
Random β=0.1	s42	0.79	0.131	0%	70.9%	29.1%
	s43	0.88	0.058	0%	93.2%	6.8%
	s44	0.89	0.070	0%	79.8%	20.2%
	Mean±std	0.85±0.06	—	0%	81.3%	18.7%

Three findings emerge:

1. **H7 is ISR-specific.** ISR reward induces H7 (conclusion = sole premise) at 64.5% across seeds; random reward produces exactly 0% H7 across all 3 seeds. H7 is a gaming strategy that directly inflates ISR: a single-premise proof where the premise equals the conclusion yields ISR≈1.0.
2. **Random reward achieves higher, more stable prove rate.** Random β=0.1 yields PR=0.85±0.06 vs. ISR β=0.1 PR=0.77±0.14. ISR-as-reward provides no prove rate advantage.
3. **ISR reshapes but does not reduce hacking.** Conclusion-in-premise rates are similar across conditions, but ISR shifts the composition from H1 (multi-premise embedding) to H7 (single-premise tautology).

4.3.2 Premise-Count Confound Correction

Initial analysis suggested that ISR seed 42 genuine samples had 2.4× higher ISR than random genuine samples. However, this difference is entirely driven by premise count: ISR-trained models produce shorter proofs, and LOO ablation on fewer premises yields mechanically higher ISR (removing 1 of 2 premises has a 50%+ chance of being "necessary").

Table 5 presents the premise-count-matched analysis.

Table 5: Premise-count-matched ISR comparison (genuine samples with ≥3 premises).

Condition	n	Mean ISR
ISR s42 genuine, $n_{\text{prem}} \geq 3$	11	0.2578
Random s42 genuine, $n_{\text{prem}} \geq 3$	21	0.2434
Ratio	—	1.06×

After controlling for premise count, the ISR advantage vanishes (1.06×, no meaningful difference). This reveals a fundamental limitation of LOO-based metrics: fewer components mechan-

ically yield higher necessity scores, independent of reasoning quality. We treat this transparent correction as a methodological contribution.

4.3.3 ISR β=0.3: Classic Goodhart Collapse

At higher reward weight, ISR-as-reward induces classic Goodhart collapse (Table 6).

Table 6: β=0.3 ISR vs. Random comparison (FOL).

Condition	PR	ISR	Status
ISR β=0.3	0.18	0.026	Goodhart collapse
Random β=0.3	0.48	0.094	Degraded but functional
Exec-only	0.93	0.132	Hacking but high PR

High-dose ISR reward is actively misleading: prove rate drops to 18%, 2.7× worse than random reward (48%). The model attempts to optimize ISR directly but produces unprovable outputs.

4.3.4 SFT Baseline: Hacking Is RL-Specific

Table 7 compares SFT against GRPO to confirm that reward hacking originates from RL optimization, not from the data or model.

Table 7: SFT vs. GRPO hacking comparison (FOL).

Method	PR	Genuine	H7	H1
SFT	0.14	92.9%	0%	7.1%
GRPO exec-only (s42)	0.93	12.9%	0%	87.1%
GRPO ISR β=0.1 (s43)	0.61	∼3%	96.7%	0%

SFT achieves low prove rate (14%) but 92.9% genuine formalizations—the model learns genuine reasoning structure from supervised signal, not hacking strategies. Proved-only ISR under SFT is 0.571, reflecting authentic component necessity from genuine reasoning (vs. artificially inflated values under GRPO with ISR reward). This confirms that Goodhart effects are RL-reward-specific, not inherent to fine-tuning.

4.3.5 Dose-Response Summary

The full β dose-response landscape (Table 8) reveals a lose-lose dynamic: low-dose ISR reward induces ISR-specific gaming (meta-Goodhart), while high-dose ISR reward collapses prove rate (classic Goodhart).

Table 8: ISR reward dose-response landscape (FOL, seed 42 where single-seed).

β	ISR PR	Random PR	ISR H7	Random H7	Interpretation
0 (exec)	0.93	—	0%	—	H1 dominates
0.1	0.77±0.14	0.85±0.06	64.5%	0%	Meta-Goodhart
0.3	0.18	0.48	—	—	Classic Goodhart

4.4 Claim 3: Preliminary Cross-Logic Evidence

4.4.1 Prolog Exec-Only: ISR Collapse Replicates

Table 9 presents the cross-logic ISR collapse comparison.

Table 9: Cross-logic exec-only ISR collapse (3 seeds each).

Logic	Seed	PR	ISR
FOL	s42	0.93	0.132
	s43	0.14	0.0
	s44	0.98	0.58
Prolog	s42	0.83	0.031
	s43	0.56	0.008
	s44	0.21	0.081
FOL mean±std		0.68±0.47	0.24±0.30
Prolog mean±std		0.53±0.31	0.04±0.04

Prolog exec-only ISR is consistently near zero across all 3 seeds (0.008–0.081), replicating the FOL ISR collapse pattern despite a different logic type and solver. Cross-seed PR instability replicates as well (Prolog σ=31pp, FOL σ=47pp). The hacking mechanism differs—Prolog models exhibit ground fact enumeration (redundant fact listing) rather than FOL's conclusion embedding—but ISR detects structural degeneration in both cases.

4.4.2 Prolog STA-as-Reward: Suggestive Goodhart Replication

Table 10 presents Prolog STA-as-reward results across 3 seeds.

Table 10: Prolog STA-as-reward 3-seed hacking composition.

Seed	PR	Proved-only ISR	H7	H1	Genuine	n_{proved}
s42	0.50	1.00	100%	0%	0%	50
s43	0.13	0.62	23%	46%	31%	13
s44	0.62	0.68	56%	21%	23%	62
Mean	0.42	—	59.7%	—	—	—

Seed 42 shows extreme Goodhart: 100% H7, training saturates by step 100, with every proved sample being a single-clause tautology. Seed 44 partially replicates this (H7=56%), with a mixture of hacking strategies. However, seed 43 diverges: training instability yields only 13% prove rate with mixed hacking modes (H7=23%, H1=46%, genuine=31%), and the small sample (n=13) limits confidence.

The 3-seed H7 mean is 59.7% (range 23%–100%), with 2/3 seeds showing H7-dominant patterns. We characterize this as *preliminary cross-logic evidence with consistent but heterogeneous hacking patterns*: H7 is consistently elevated relative to baseline, but cross-seed variation is substantial. Prolog's simpler semantics may facilitate faster and more complete gaming (s42 saturates at step 100 vs. FOL needing 300+ steps), while also producing more heterogeneous failure modes across seeds.

4.5 Additional Experiments

4.5.1 $O(1)$ vs. $O(N)$ Ablation

We evaluate a fast $O(1)$ approximation of STA that tests a single randomly sampled component instead of all n components (Table 11).

Table 11: $O(1)$ approximation vs. full $O(N)$ LOO hacking detection recall.

Hacking Mode	$O(1)$ Recall	$O(N)$ Recall
H7 (tautological)	0.85	1.00
H1 (conclusion embed)	0.30	1.00
H3 (complexity)	0.15	1.00

The $O(1)$ approximation achieves Spearman ρ=0.35 against full LOO ISR and detects H7 adequately (recall 0.85, due to the obvious 2-component structure) but misses subtle H1 and H3 patterns (recall 0.30 and 0.15). The overall false negative rate is 55%. We conclude that full $O(N)$ LOO is necessary for reliable hacking diagnosis; the cost is justified by the diagnostic quality.

4.5.2 NLI Downstream Validation (Negative Result)

We evaluate whether ISR predicts downstream NLI accuracy to test whether structural integrity translates to semantic faithfulness. Overall NLI accuracy is 34.1%, with Spearman ρ(ISR, NLI accuracy) = -0.049 (p=0.187, not significant). This negative result precisely delineates ISR's capability boundary: ISR measures *syntactic component necessity*, not semantic faithfulness. High ISR indicates that each component is needed for the proof to succeed, but does not guarantee that the formalization faithfully represents the original natural language meaning.

4.5.3 Historical: Convex Combination Formula

An earlier reward formulation using convex combination ($r = \alpha \cdot r_{\text{exec}} + \beta \cdot r_{\text{exec}} \cdot \text{ISR}$, $\alpha + \beta = 1$) was abandoned because the constraint $\alpha + \beta = 1$ means that increasing ISR weight necessarily decreases execution weight, making the ISR term a penalty when ISR < 1. This experiment (PR=77.3%, ISR/PR ratio=0.298) motivated the switch to the multiplicative bonus formulation (Eq. 4). The corresponding random convex control collapsed to 100% tautological proofs. We note this as a design decision; the convex and multiplicative formulations are structurally different and should not be directly compared.

5 Discussion

5.1 ISR's Fundamental Asymmetry

Our central finding is a fundamental asymmetry in ISR's utility: it is reliable as a diagnostic observer but unreliable as an optimization target. As a diagnostic, ISR reveals structural degeneration that prove rate conceals—the 0.64→0.003 collapse trajectory is invisible to PR (which rises monotonically). As a reward, ISR induces meta-level Goodhart effects: models learn to inflate ISR through conclusion duplication rather than genuine structural improvement.

This asymmetry has implications beyond our specific setting. Any proxy metric considered for reward shaping should be stress-tested for gaming before deployment. The meta-Goodhart pattern—where a diagnostic metric intended to detect gaming itself becomes a gaming target—may apply to other structural quality metrics in RL-trained systems.

5.2 Premise-Count Bias in LOO Metrics

The premise-count confound (Section 4.3.2) reveals a general limitation of LOO-based metrics: component necessity scales inversely with component count. A 2-premise formalization where both premises are needed achieves ISR=1.0 trivially, while a 10-premise formalization requires all 10 premises to be individually necessary for the same score. This mathematical property means ISR rewards brevity independent of reasoning quality—a form of metric bias that practitioners should account for when using LOO-based diagnostics.

5.3 Cross-Logic Generality and Limitations

The exec-only ISR collapse replicates consistently across FOL and Prolog (Section 4.4), supporting diagnostic transfer. However, the STA-as-reward results are heterogeneous: Prolog seed 42 shows extreme Goodhart (100% H7), while seed 43 is anomalous (mixed hacking, training instability, n=13). We attribute this heterogeneity to Prolog's simpler semantics, which create a sharper reward landscape where seeds either converge cleanly to tautological collapse or fail to converge.

Limitations. (1) All experiments use $\sim$1.5B parameter models; scaling behavior is unknown. (2) Only 2 logic types are tested; additional formal languages (e.g., SMT-LIB) would strengthen generality claims. (3) Three seeds, while revealing cross-seed instability as a finding, limit statistical power. (4) ISR measures syntactic necessity, not semantic faithfulness (ρ=$-$0.049 with NLI accuracy). (5) We do not propose a positive training method; our contribution is diagnostic.

6 Conclusion

We introduced Solver Trace Ablation (STA), a diagnostic protocol for detecting reward hacking in GRPO-trained neurosymbolic formalizers via leave-one-out component necessity testing. Through systematic experiments across FOL and Prolog with 3 seeds and up to 1000 training steps, we demonstrated three principal findings:

1. ISR detects structural degeneration invisible to prove rate, with a collapse trajectory from 0.64 to 0.003 over training, accompanied by a 6-mode hacking taxonomy.
2. Using ISR as a training reward induces meta-level Goodhart's Law—the diagnostic itself becomes a gaming target—with no genuine structural advantage after controlling for premise count (1.06$\times$).
3. ISR collapse under execution-only training replicates across logic types, providing preliminary cross-logic diagnostic transfer evidence.

Our findings yield actionable guidance: use ISR with component count as a joint post-hoc diagnostic, never as a training reward; distrust single-seed GRPO results; and stress-test any proxy metric for gaming before using it as an optimization signal.